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To cite this article: Qicheng Xu *et al* 2025 *J. Phys.: Conf. Ser.* **3109** 012095

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A Method for Constructing Space Target Satellite Image NeRF Dataset Based on Data Generation

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Abstract. The importance and significance of datasets in the 3D reconstruction of space targets lie in their provision of rich information from real-world scenarios, including images, point clouds, depth information, and more. These data can help algorithms understand the spatial structure, geometric features, and surface textures of target objects, providing crucial inputs for reconstruction and modelling. Thoughtful selection and processing of datasets can enhance the accuracy and efficiency of reconstruction algorithms, thereby driving the development and application of 3D reconstruction techniques for space targets. Therefore, for 3D reconstruction of space targets, establishing high-quality and diverse datasets is crucial as they directly impact the quality and reliability of reconstruction results. This paper primarily focuses on the study of the simulation generation method of space target satellite-based sequence images based on Kubric. Typical three-dimensional models of space targets are established, their optical characteristics are analysed in depth, and the imaging characteristics of typical space targets are systematically summarized and generalized. Based on the research of the imaging characteristics of space target satellites, using the Kubric simulation library, the simulation of space target imaging is achieved through processes such as designing satellite imaging scene scripts, setting lighting conditions, and adding motion blur effects, successfully generating space target image simulation data with realistic physical world effects. Based on this generation method, a dataset of simulated space target satellite-based sequence images under multi-perspective imaging conditions is constructed, providing data support for space three-dimensional visual perception tasks.

Keywords: Space target; data generation; sequential images; 3D reconstruction; neural rendering

1. Introduction

Data generation has wide applications in many areas, including but not limited to image generation, text generation, audio generation, and more. In many cases, it may not be feasible to acquire enough real data for model training, making data generation particularly crucial. For example, in the field of computer vision, a large number of annotated images may be required to

train models[1]. However, acquiring and annotating these images can be both expensive and time-consuming. Through data generation, a plethora of synthetic images can be produced, automatically annotated, significantly enhancing the efficiency of data acquisition[2]. Furthermore, data generation can be employed for data augmentation, expanding the training set by generating new and diverse training samples to improve the model's generalization capability.

Despite the emergence of some public datasets for space targets[3-5, 7], there remains a substantial gap in both quantity and quality compared to fields like facial recognition and natural scenes. The current research faces challenges due to the lack of space target datasets, including difficulties in data acquisition in space environments, issues with data accessibility and sharing, and the diversity of space targets. Data collection in space requires precise satellites or spacecraft, leading to high costs and complexity that limit dataset growth. Moreover, restrictions in sensitive areas like military and commercial sectors hinder data sharing. The wide range of space targets, including artificial satellites, space debris, and asteroids, each with distinct characteristics, presents significant challenges in constructing a diverse and representative dataset.

The research focuses on the significance of generating satellite space target images in various aspects. Firstly, through generation techniques, it enables the simulation and prediction of different states of satellite space targets, essential for space mission planning and tasks like target identification. Secondly, image generation helps in data augmentation by creating synthetic images when real data is limited, enhancing model training and aiding in tasks like 3D reconstruction. Additionally, the generated images can be used to evaluate and compare different recognition algorithms for space targets. Lastly, these images have educational value and can be used for public outreach, improving understanding and recognition of satellite space targets.

Therefore, current research requires further efforts to establish rich, high-quality space target datasets, enhancing research levels in fields such as space target recognition, tracking, and motion prediction. This will provide more complete data support and assurance for the development and application of relevant fields. Research on generating satellite space target images holds important theoretical and practical significance. In summary, our key contributions are:

We propose a space target sequence image generation technique based on the Kubric data generation pipeline. This technique not only allows for fine and complex control of data but also significantly improves the flexibility and efficiency of data processing. The introduction of this technique effectively alleviates the current shortage of space target sequence image datasets, providing valuable data support for research in related fields. Experimental results demonstrate that the generated image data exhibit high realism, accurately reproducing the actual state of space targets in the space environment, thereby offering strong data support for research on space target identification and tracking.

2. Related work

The Spacecraft PosE Estimation Dataset (SPEED)[4] for spacecraft attitude estimation challenges is jointly provided by the European Space Agency and Stanford University. This dataset aims to support tasks related to space target attitude estimation, including simulated images of the Tango satellite and real images captured by the TRON device in the SLAB laboratory. The SPEED dataset consists of approximately 15,000 images with a resolution of

1920×1200, annotated with quaternions and 3D depth distances. In 2021, SPEED+[3] was introduced as the next-generation dataset to address cross-domain differences in spacecraft attitude estimation, comprising 60,000 synthetic images and 9,531 real images from TRON. To further enhance the 6-D attitude estimation task of space targets, Hu et al. provided the Swiss Cube dataset[5], featuring satellites captured in a polar orbit at a distance of 700 km. Proença and Gao[6] utilized Unreal Engine 4 for data rendering to generate images based on the Alliance and Dragon spacecraft models. It is noteworthy that the SPEED, URSO, and Swiss Cube datasets focus on the 6-D attitude estimation of space targets, utilizing one or multiple satellite models for simulation.

In addressing the space target classification task, Liu[7] constructed a space target dataset using the STK tool. They utilized a series of motion blur and defocus blur image processing algorithms to simulate real imaging conditions in space. This dataset includes 100 images across 4 categories. Zhang[8] established the BUAA-SID1.0 space target image dataset. They imported 20 different types, shapes, and functional satellite models into the 3dsMax software. By sampling around the satellite models at regular intervals and utilizing single lighting from the external environment, over 4,000 images were generated in total. Due to limitations in model accuracy and texture in STK, the final simulation effect of this database was relatively poor, lacking background environment simulation.

During the 2021 IEEE International Conference on Image Processing, the SPARK dataset[9] was provided by the Center for Security, Reliability, and Trust of the Interdisciplinary Center for Utilizing Space Environment Knowledge in the First SPACecraft Recognition Competition. This dataset is used for tasks such as space target classification and detection. In the SPARK dataset, 10 different real satellite and spacecraft models (including AcrimSat, Aquarius, Aura, Calipso, CloudSat, Jason, Terra, TRMM, Sentinel6, and 1U generic cube satellite) and 5 fragment models (including external storage tanks for spacecraft, orbital docking systems, communication panels, thermal protection tiles, and connection rings) are utilized to simulate space target images. This dataset enhances the fidelity of simulated data by adding high-resolution earth images to the images. Additionally, besides tasks such as attitude estimation and space target classification, satellite structure detection and segmentation datasets are also valuable. Dung[10] provided a dataset for detection, image segmentation, and structure recognition using a combination of segmentation networks and manual post-processing. However, these images are primarily from artistic rendering and cannot fully depict the environments in which space targets are located.

Neural Radiance Fields (NeRF)[11], as an advanced view synthesis technology based on deep neural networks, combines Multilayer Perceptron (MLP) and implicit neural scene representations to achieve fine rendering of object representations, thereby generating high-quality view synthesis effects[12-14]. Compared to traditional 3D reconstruction methods, NeRF does not require explicit depth information or complex geometric models. Instead, it learns the scene representation directly from multi-view images and camera poses, making it highly suitable for tasks such as space target reconstruction. However, NeRF's performance heavily relies on the quality and diversity of the input images, which is where our proposed method based on Kubric comes into play. By using Kubric to generate high-quality synthetic images with accurate camera poses, we can effectively train NeRF models for space target reconstruction. This combination of Kubric and NeRF offers a significant advantage over traditional methods, which often struggle with the limited availability of real-world space target images.

Compared to traditional 3D neural representations or view synthesis methods, NeRF model's significant advantage lies in its reliance solely on learning the scene from multi-view

images and camera pose information, without requiring explicit supervision of the three-dimensional depth of the images. This characteristic allows NeRF to achieve significant improvements in visual quality expression compared to many traditional view synthesis methods and other advanced technologies[15]. Subsequent improvements and optimizations tailored to NeRF have further demonstrated its outstanding performance and application potential.

3. Materials and Methods

3.1. Analysis of Space Environment and Space Target Characteristics

In the space environment, space targets are influenced by space perturbations, making it difficult to accurately describe their orbits. Furthermore, obtaining surface material and texture details of space targets manually is challenging. These environmental factors limit the authenticity of simulating space target datasets. Therefore, this section decomposes the space model into the space environment, including lighting models, 3D models of space targets, Earth models, and cameras.

Space target models include satellites and space debris. Space debris refers to the debris and fragments of all man-made objects in orbit or out of orbit, which may result from spacecraft decommissioning, collisions, explosions, or other types of events. Additionally, it includes some natural objects in orbit. According to current statistics from the Space-Track website[16] as of March 10, 2024, excluding military satellites from various countries, there are approximately 28,950 operational satellites in orbit, around 6,560 rocket bodies, and approximately 34,770 pieces of space debris.

In space, visible light primarily comes from the sunlight emitted by the sun and sunlight reflected by the Earth. Within the visible light spectrum, the sun serves as the main radiation source, and the background radiation from other stars can be disregarded for targets[17]. Space targets facing the Earthward side receive some sunlight reflected by the Earth, which is weaker in effect compared to direct sunlight from the sun. To simulate real sunlight, only one directional light source is added to the model in this section. The key in the study is to calculate the energy of sunlight reflected by the target, which can be accurately computed by considering parameters such as the solar radius, Earth-sun distance, using principles of infrared physics and the Planck formula[18]. The figure illustrates the optical background of space targets, with E_s representing direct sunlight and E_c representing sunlight reflected by the Earth and its atmosphere.

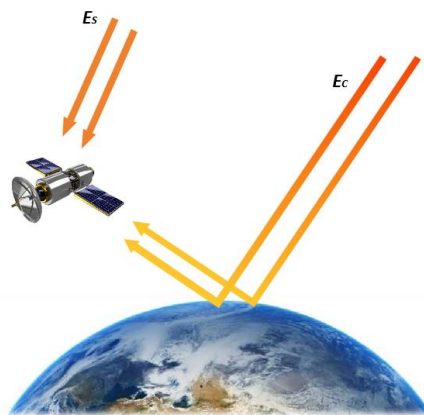


Figure 1. Optical background of space targets showing direct sunlight (E_s) and Earth-reflected sunlight (E_c).

Between 1970 and 1981, NASA used high-precision measurements from a series of satellites and other spacecraft to determine the value of the solar constant in 1982[19]:

$$E_s = 1371 \pm 5 W / m^2 \quad (1)$$

Equations (2) and (3) can be used to calculate the irradiance of the sun in the visible light spectrum:

$$E_{vis} = E_s \cdot k_{\lambda_1-\lambda_2} \quad (2)$$

$$k_{\lambda_1-\lambda_2} = \frac{\int_{\lambda_2}^{\lambda_1} \frac{c_1}{\lambda^5} \cdot \frac{1}{e^{c_2/\lambda T} - 1} d\lambda}{\sigma \cdot T^4} \quad (3)$$

Here, $k_{\lambda_1-\lambda_2}$ is defined as the ratio of the energy of a black body at temperature (T) in a specific wavelength band $\lambda_1-\lambda_2$ to the total energy over all wavelengths. Taking the imaging band as $\lambda_1=0.45\mu m, \lambda_2=0.90\mu m$, we have $k_{0.45-0.90}=0.4749$.

Therefore, the radiation constant of the sun in the visible light spectrum can be calculated as:

$$E_{vis} = 651.0978 W / m^2 \quad (4)$$

Brightness calculations are influenced by the detector equipment; the irradiance of the target at a certain distance remains consistent, but the brightness measured by different detectors depends on the efficiency of the optical system, circuitry, and device performance.

In astronomy, magnitude is a standard used to measure the brightness of celestial objects in the sky. Brighter celestial objects have smaller magnitude values, while dimmer celestial objects have larger magnitude values. Magnitudes are typically categorized into apparent magnitude and absolute magnitude. Apparent magnitude represents the brightness of an object as observed from Earth, while absolute magnitude indicates the actual brightness of an object at a fixed distance. Magnitudes are measured on a logarithmic scale, where each increase in magnitude represents an exponential increase in brightness; specifically, an increase of one magnitude corresponds to a brightness increase of approximately 2.512 times[20]. This logarithmic scale design enables us to accurately compare the brightness of different stars or celestial objects, leading to a better understanding of their characteristics.

The formula for calculating the apparent magnitude of a space target is:

$$m = 17.5 - 2.5 \log_{10} \left(\frac{E_{vis}}{\pi} \cdot \rho \cdot \cos \alpha \cdot \Omega \cdot \frac{1}{\Delta B} \cdot \frac{R}{A \cdot \eta_e} \right) \quad (5)$$

From equation (5), it is evident that the apparent magnitude of the target is related to the parameters of the detection system, solar constant, orbital parameters, and the surface material of the target. Specific parameters are detailed in the table below.

By calculating the relevant formulas for solar radiation to obtain the solar radiation constant, we apply it to the parameter settings of simulating solar illumination intensity in the Kubric scene. This is done to more realistically reflect the illumination conditions of space targets in space. Based on the characteristics of the surface materials of space targets and the results of illumination calculations, we reasonably adjust the material properties of objects in Kubric, such as reflectivity and roughness, to make the generated images more consistent with the actual situation. In addition, the orbital parameters and motion characteristics of space targets provide a basis for setting the motion path and shooting angle of the camera, ensuring

that the multi - perspective images obtained can comprehensively reflect the characteristics of space targets. These formulas and characteristic analysis results together form the basis of the subsequent method for constructing space target datasets using Kubric and Colmap, and are closely related to the entire research method process.

Table 1. Parameters affecting the apparent magnitude of space targets.

Detection system	Ω	Instantaneous field of view of the detector
	ΔB	Bandwidth
	R	Response rate of the detector
	A	Area of the detector unit
	η_e	Detector electron conversion efficiency
Solar constant	E_{vis}	Solar irradiance in the visible light spectrum
Orbital parameters	α	Solar incidence angle
Surface material	ρ	Space target diffuse reflectance (in the visible light range, wavelength-independent)

In general, space target images have the following characteristics:

1. Mostly black and white images, lacking color information.
2. Space is a dark environment, resulting in low image brightness.
3. Spacecraft surfaces are predominantly made of metallic materials, leading to significant surface brightness variations and encountering the “sun glint” issue.
4. Space can be approximated as having only the sun as a light source, creating unidirectional lighting and the “halo” issue where some angles lack information.
5. Images of space-based satellites may exhibit motion blur due to the spacecraft’s orbital movement and camera shake.

Considering these characteristics, this chapter generated simulated sequential images of space targets.

3.2. Methodology for Generating Sequential Images of Space Targets

3.2.1. Introduction to the Kubric Library

Kubric[21], a tool for visualization and rendering, encompasses eight essential concepts - scenes, objects, lights, cameras, materials, animations, rendering, and scripting - that are intricately interconnected to enable a comprehensive and efficient visualization and rendering workflow.

As the fundamental unit in Kubric, a scene serves as the container for all 3D objects, lights, cameras, and other attributes. Multiple scenes can be crafted, each with its own distinct set of elements and parameters, providing the flexibility to create diverse virtual environments. Objects, which include 3D models, geometric shapes, and particle systems, are the visual components within a scene. They can be added, edited, or removed from the scene, and their properties such as position, rotation, and scale can be adjusted, thereby influencing the overall appearance and layout of the scene.

Lights, which come in various types like point lights, directional lights, and spotlights, are crucial for illuminating the scene. By modifying light properties such as position, color, and

intensity, users can achieve the desired lighting effects, which in turn impact how objects are perceived within the scene. The camera, on the other hand, determines the observer's view of the scene. By setting properties like position, direction, and field of view, it controls the scene's rendering view, allowing users to focus on specific areas or capture different perspectives.

Materials play a significant role in defining the appearance and surface characteristics of objects. With options for color, texture, and reflectance, different materials can be assigned to objects to achieve specific visual effects, enhancing the realism and visual appeal of the scene. Kubric also supports animation effects for both objects and cameras. By setting keyframes on the timeline and leveraging interpolation, smooth and natural - looking animations can be created, adding a dynamic element to the scene.

When it comes to the final output, Kubric allows scenes to be rendered as either images or videos. By selecting appropriate rendering settings such as resolution, frame rate, and renderer, high - quality visual results can be obtained. Additionally, Kubric provides a Python API, enabling users to create and control scenes programmatically. This programmability allows for the customization of the scene creation, editing, and rendering processes through scripts, making Kubric highly adaptable to different requirements.

Due to its integrated features, Kubric is well - suited for a wide range of applications across fields like movie production, game development, and scientific visualization. It serves as a powerful tool for achieving diverse visualization goals, whether it's creating realistic virtual worlds, producing engaging animations, or presenting complex data in an intuitive manner.

3.2.2. Workflow for Generating Sequential Images of Space Targets

The process of generating sequential images of space targets involves a series of interconnected steps. First, the scene is set up. This includes carefully choosing the background, light sources, and camera positions. The background and lighting are crucial for simulating the space environment accurately, and the camera position determines the perspective from which the space targets will be viewed. After the scene is established, objects within the scene are created and edited. Kubric offers a diverse library of 3D objects, featuring a wide variety of shapes, models, and geometries. Existing 3D satellite models are utilized and modified through operations like scaling, rotating, and moving to fit the specific requirements of the space target simulation.

Following object creation and editing, materials and textures are applied to the objects. By adding colors, textures, and reflections, the appearance and realism of the objects are significantly enhanced, making them more representative of real - world space targets. Additionally, objects can be equipped with animation paths, keyframes, and various animation effects. Effects such as particle effects and lighting effects are also added to further enrich the scene. In this particular process, a unidirectional light source is used to mimic sunlight in space, and the background is set to a dark color to simulate the vastness of the space environment. The algorithmic structure underlying these operations is illustrated in Figure 2.

Once the scene setup, object editing, and effect addition are completed, the rendering and output stage begins. Kubric leverages advanced rendering and ray - tracing technology to generate high - quality images and animations. Users can define the rendering resolution and format according to their needs, and the final results can be saved as images, videos, or other file types. To more realistically simulate the scenes captured by satellites in high - speed motion, considering factors like camera shake, motion blur is added to the generated images.

This comprehensive workflow encompasses scene setup, object creation and editing, material and texture settings, animation and effect additions, as well as rendering and output.

By going through these steps in sequence, highly realistic virtual scenes and images can be created, effectively meeting the requirements for generating sequential images of space targets. The overall workflow is visually presented in Figure 3 for a better understanding.

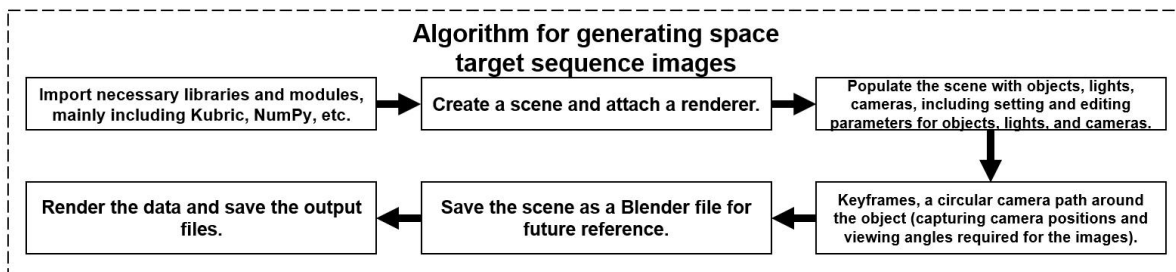


Figure 2. Algorithm structure for generating sequential images of space targets.

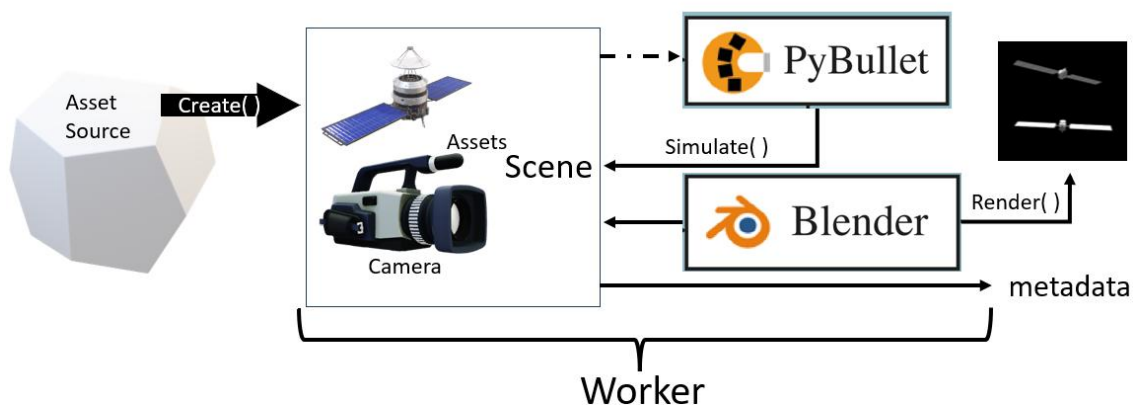


Figure 3. Kubric working principle.

3.2.3. Configuration of Pose Information for Sequential Images of Space Targets

When training and rendering with NeRF, a series of dataset requirements need to be met. Firstly, the dataset should contain multi-view images to capture various aspects of the scene from different angles. Each image should also include camera parameters, such as the intrinsic and extrinsic matrices, to determine the projection location of the scene. Depth information or camera distance information is also necessary to accurately estimate the geometric shape and resolution of the scene. Additionally, the positions and orientations of cameras in the dataset should cover the entire scene to ensure NeRF can comprehensively capture the scene. Using high-resolution images helps enhance the rendering quality and accuracy.

The dataset should be diverse and dense, comprising a variety of rich scenes and ensuring a sufficiently high density of camera viewpoints for NeRF to effectively capture scene details. Ideally, selecting datasets of real-world scenes aids the model in learning the complex lighting and geometric features of real scenes. Therefore, when selecting datasets for NeRF, a comprehensive consideration of dataset diversity, density, and realism is required to ensure that the trained model can accurately capture and render the details and features of complex scenes.

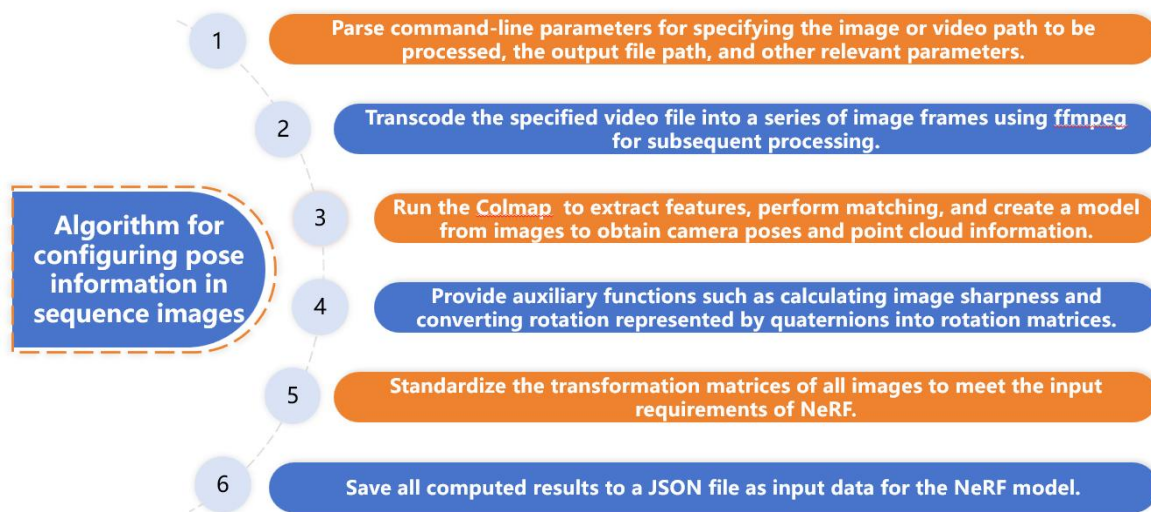


Figure 4. Flowchart for Configuring Pose Information in Sequential Image.

```

import kubric as kb
import numpy as np
from colmap_wrapper import run_colmap # Assuming this wrapper function exists to call Colmap

# Create a scene
scene = kb.Scene()
# Set the background to black (simulating space)
scene.background = kb.Color(0, 0, 0)
# Set a point light to simulate the sun, with an intensity of 20
light = kb.PointLight(position=[0, 0, 100 * satellite_model_size], intensity=20)
scene.add(light)
# Set up the satellite model (assuming satellite model data is available)
satellite = kb.Asset(filename='satellite_model.obj')
scene.add(satellite)
# Set up camera parameters
camera = kb.PerspectiveCamera()
camera.position = [0, 100, 0]
camera.look_at([0, 0, 0])
scene.add(camera)
# Set up animation (assuming the satellite rotates around an axis)
animation_length = 100
for frame in range(animation_length):
    satellite.rotation = [0, np.radians(frame * 360 / animation_length), 0]
    scene.set_frame(frame)
# Render the scene
renderer = kb.renderer.Blender(scene)
images = []
for frame in range(animation_length):
    image = renderer.render_still(frame=frame)
    images.append(image)

```

Figure 5 Image generation algorithm pseudo code

After obtaining simulated sequential images of space targets, a series of operations should be conducted to acquire the positional information of the images. Figure 4 illustrates the process for configuring pose information in sequential images. Colmap[22] is an open-source computer vision software used to generate 3D sparse point clouds and camera pose information. Colmap employs multiple key technical principles in camera pose estimation. Firstly, significant feature points are extracted from the input images using methods like SIFT, ORB, etc., for subsequent

feature matching. Then, Colmap computes feature point descriptors and performs feature matching to discover shared feature point pairs between different images. Utilizing the feature point matching results, algorithms such as RANSAC are applied to estimate the fundamental matrix, describing the projection relationship between different images and deriving the relative camera poses. Once the relative poses are obtained, Colmap triangulates the three-dimensional coordinates of the feature points using the triangulation technique. Furthermore, global optimization is conducted through bundle adjustment to maximize matching observation data and minimize reprojection errors, aiming to improve the accuracy of camera poses and 3D points. By optimizing reprojection errors to adjust camera poses and 3D point positions, the overall reconstruction quality is enhanced. In summary, Colmap integrates various key technologies, including feature point extraction, matching, fundamental matrix estimation, triangulation, and bundle adjustment, to obtain accurate camera poses and 3D scene structure information through the optimization process, achieving 3D reconstruction of image sequences. The schematic illustration of the NeRF reconstruction process after obtaining pose information is shown in Figure 6. From the figure, it can be observed that the images containing poses are distributed around the central object, with each pixel in the image emitting rays intersecting at the location of the central object to reconstruct the 3D model. The diagram also illustrates the significant impact of the accuracy of image poses on the reconstruction quality.

In this study, we leverage the powerful view synthesis capabilities of NeRF (Neural Radiance Fields) to perform 3D reconstruction on the space target sequence images generated based on Kubric. NeRF can implicitly represent the geometric structure and radiance field of a scene by learning multi-view images and camera pose information, thus achieving high-quality view synthesis and 3D reconstruction effects.

Specifically, we use Kubric to generate space target sequence images with different viewpoints, lighting conditions, and target motion states. Then, we utilize Colmap to obtain the camera poses and point cloud information of these images, providing accurate input data for the NeRF model. By training the NeRF model, it can learn the appearance features and geometric structures of space targets from different viewpoints, enabling precise 3D reconstruction of space targets.

Different from other NeRF-based studies, we focus on the imaging simulation and 3D reconstruction of space targets in complex space environments. By optimizing the scene setup and parameter adjustment of Kubric, we generate image data that is more consistent with the actual space environment, improving the accuracy and robustness of NeRF in the 3D reconstruction task of space targets.

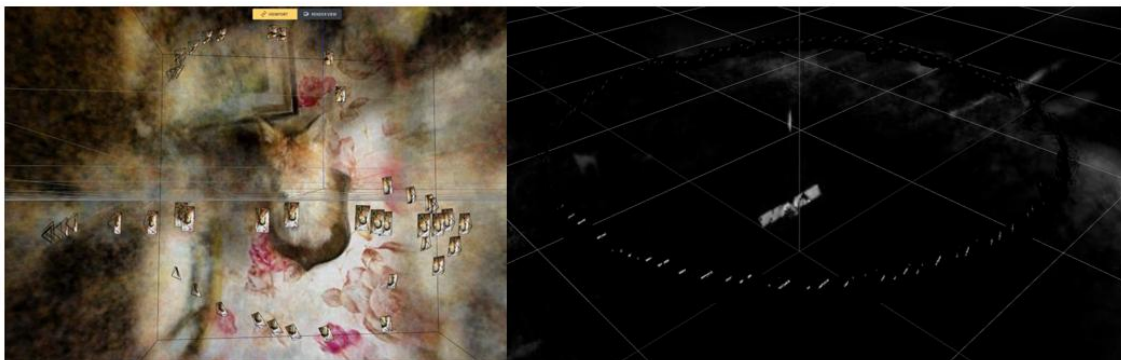


Figure 6. Schematic Diagram of Reconstructing Sequence Images with Pose Information.

4. Results and Discussion

Following the workflow outlined in the previous section, we configured the scene with a dark background to simulate the space environment. The sun was modeled as a point light source with an intensity of 20, positioned at a distance 100 times the size of the satellite's 3D model to replicate solar lighting conditions. The camera path was designed as a circular trajectory to capture multi-angle images of the space targets. To reflect real-world imaging conditions, the generated images were set to grayscale, and random motion blur was introduced to simulate the effects of high-speed satellite motion and camera shake. The brightness of the images was moderately enhanced for better visualization, as the original lighting conditions resulted in low brightness.

The generated image samples demonstrate high realism, accurately capturing the optical characteristics of space targets. Figure 7 shows a simulated image of a spacecraft under unidirectional lighting, with clear surface details and motion blur effects. Figure 8 and Figure 9 present additional examples, highlighting the diversity of lighting conditions and camera angles. These results indicate that our method is capable of generating high-quality images suitable for space target identification and tracking tasks.

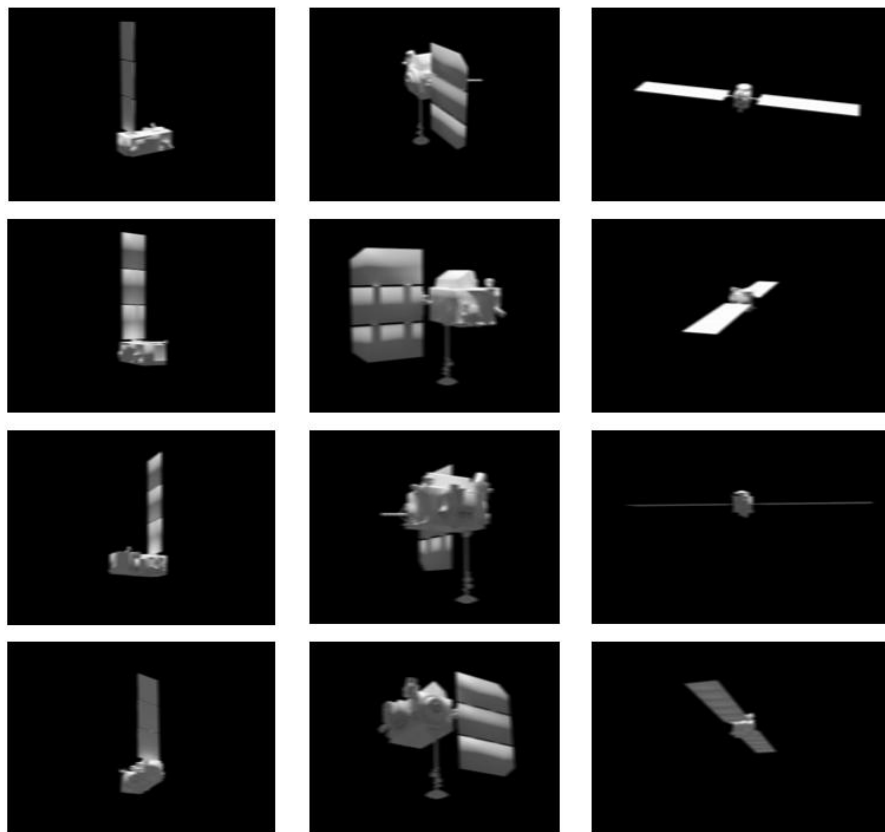


Figure 7. Simulated Image of the Spacecraft (1).

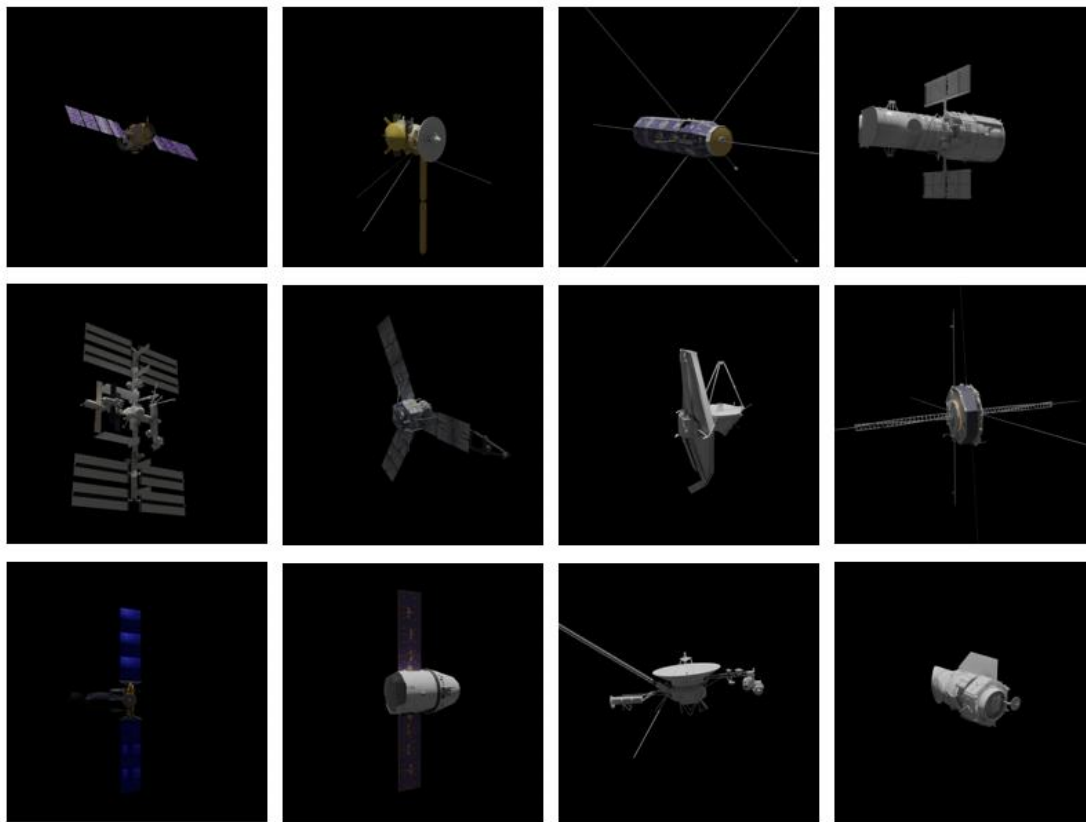


Figure 8. Simulated Image of the Spacecraft (2).

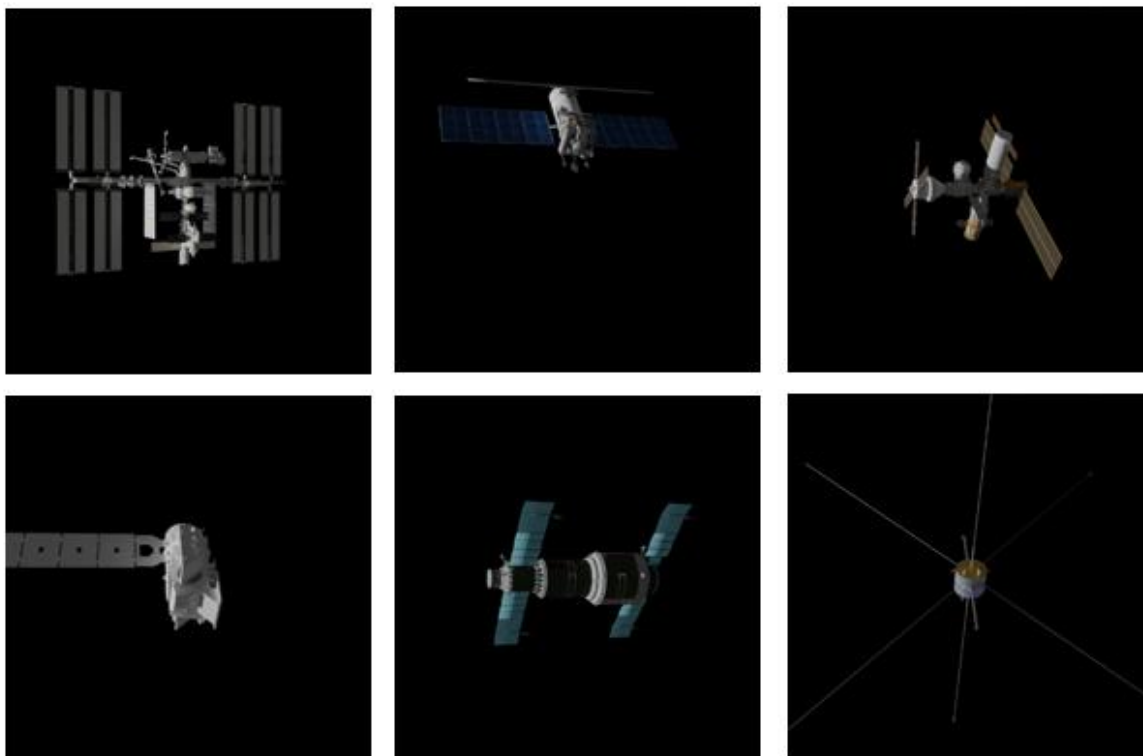


Figure 9. Simulated Image of the Spacecraft (3).

Our method offers significant flexibility in scene customization, allowing users to add new objects, modify materials, and adjust lighting configurations through an expandable scene construction module. The ability to manually script scene configurations provides a high degree of control, making the process both flexible and user-friendly. Additionally, Kubric's extensive library of filters and effects enables the creation of diverse image styles and visual effects, enhancing the realism and immersion of the generated images.

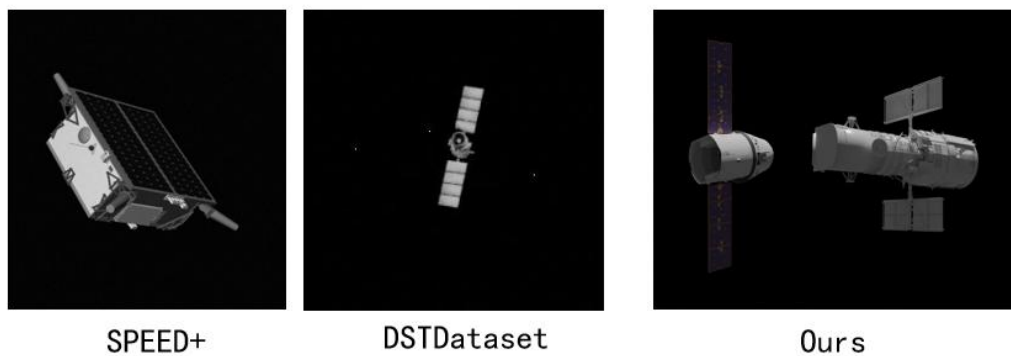


Figure 10 Comparison of Space Target Image Data

Presently, there exist two space target datasets with relatively high quality and strong realism. One is the SPEED dataset proposed by Stanford University, which was initially introduced in 2020, and its next-generation work was released in 2022. The other is a dataset published by Beijing Institute of Technology in 2022.

The SPEED+ dataset encompasses approximately 70,000 images. However, it features only one target as shown in the figure. It consists of three folders, where the first two contain photos captured under light box conditions and simulated sunlight conditions, respectively, along with synthetic data. As shown in Figure 10, the synthetic data from the SPEED+ dataset is compared with our proposed dataset. In contrast, the DST dataset incorporates 15 satellite targets and 35 space debris targets, totaling around 20,000 images, yet its resolution is relatively low.

Both of these datasets entail substantial workloads and involve extensive post-processing efforts. Nevertheless, by employing our methodology, under the hardware conditions of a laboratory desktop computer, generating 60,000 images of comparable quality to the SPEED+ dataset approximately requires two days, while producing 20,000 images similar in quality to the DST dataset takes roughly half a day. Moreover, our approach obviates the need for manual post-processing and can flexibly incorporate multiple spacecraft targets, diverse lighting conditions, and camera paths.

Compared to existing datasets such as SPEED+ and DST, our approach demonstrates several advantages. First, the ability to generate large-scale datasets with high resolution and diverse lighting conditions significantly reduces the time and effort required for manual data collection and annotation. Second, the inclusion of motion blur and grayscale effects enhances the realism of the images, making them more suitable for training machine learning models for space target identification and tracking. Finally, the flexibility of our method allows for the incorporation of multiple spacecraft targets, enabling the creation of more comprehensive datasets.

Table 2. Explanation of the Space Target Simulation Dataset.

Dataset	Ours
Number of scenes	17
Number of scene views	90

View rotation angle	360°
Lighting conditions	Sun parallel light, unidirectional light source
Background	Outer space
Types of included targets	Satellite, space station, space telescope, space probe
Image resolution	640×640
Physics simulator	PyBullet
Renderer	Blender

Additionally, this method has data augmentation capabilities that can generate more diverse and varied image data for training machine learning models, thereby improving the robustness and generalization ability of the models. Kubric's rendering engine boasts high performance and fast rendering speeds, enabling the rapid generation of high-quality images. This is highly beneficial for applications requiring the processing of a large volume of image data or those with time constraints.

At the current stage, there are still some shortcomings in image generation data in terms of detailed representation. For instance, the simulation of spacecraft surface materials is not yet fine-grained, and some non-Lambertian properties are not effectively depicted.

5. Conclusions

This research primarily focuses on the study of the simulation generation method of space target satellite-based sequence images based on Kubric. Typical three-dimensional models of space targets are established, their optical characteristics are analyzed in depth, and the imaging characteristics of typical space targets are systematically summarized and generalized. Based on the research of the imaging characteristics of space target satellites, using the Kubric simulation library, the simulation of space target imaging is achieved through processes such as designing satellite imaging scene scripts, setting lighting conditions, and adding motion blur effects, successfully generating space target image simulation data with realistic physical world effects. Based on this generation method, a dataset of simulated space target satellite-based sequence images under multi-perspective imaging conditions is constructed, providing data support for space three-dimensional visual perception tasks.

While the method of generating space target sequence images based on Kubric has achieved some encouraging results, there are still challenges and room for improvement. In future research, the following directions can be considered:

First, further optimize parameter settings and algorithm design. In this study, efforts were made to select appropriate parameter settings to generate target sequence images. However, there are still many parameters that can be further adjusted and optimized. Exploring more combinations of backgrounds, lighting, and effects can enhance the quality and diversity of generated images.

Second, increase the scale and diversity of the dataset. A relatively small and limited dataset was used in the current experiments. In future research, more samples can be collected, including a greater variety of space targets, to better reflect the diversity of real-world scenarios. This will further enhance the quality and realism of generated images.

Furthermore, conducting in-depth comparisons with other generation models is also one of the future research directions. Compared to the latest generation models such as GANs and VAEs, there is still room for improvement in the method based on Kubric. By comparing with these

advanced models, the strengths and weaknesses of the method in generating space target sequence images can be better evaluated.

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